P2 Chapter 8: Parametric Equations

Curve Sketching

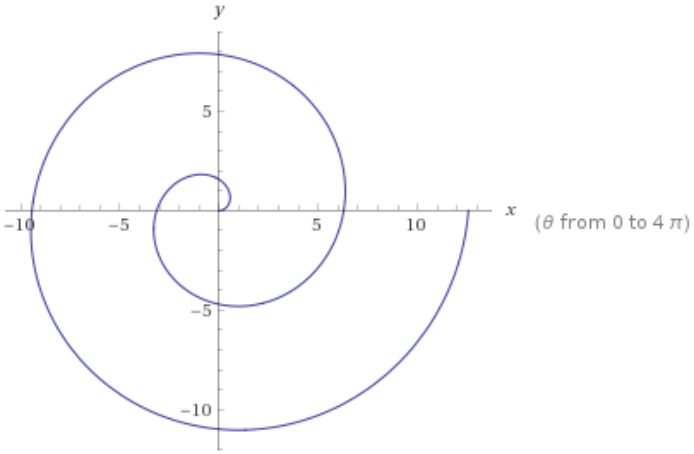
Sketching Parametric Curves

We saw that one strategy for sketching parametric curves is to convert into a Cartesian equation, and hope this is a form we recognise (e.g. quadratic or equation of circle) to appropriately sketch.
However, some parametric equations can't easily be turned into Cartesian form:

Input interpretation:

plot $x = \theta\cos(\theta)$, $y = \theta\sin(\theta)$ $\theta = 0$ to 4π

Parametric plot:



These parametric equations in Cartesian form would be $\sqrt{x^2+y^2} = \arccos\left(\frac{x}{\sqrt{x^2+y^2}}\right)$; this would obviously be incredibly hard to sketch!

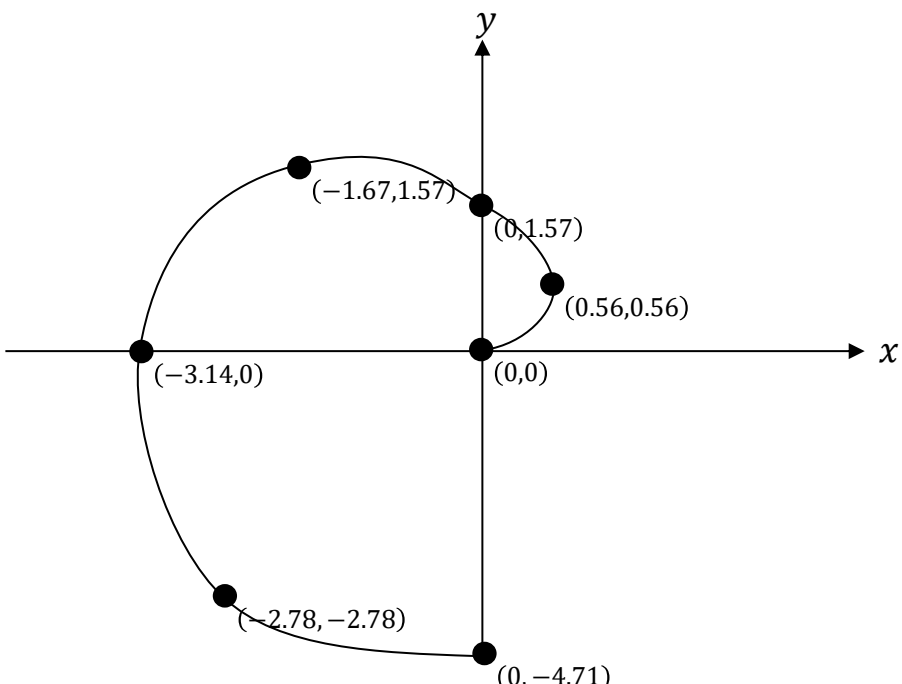
Instead we can try different values of t and determine the point (x, y) for each value to get a sequence of points…

Sketching Parametric Curves

Input interpretation:

plot $x = \theta\cos(\theta)$, $y = \theta\sin(\theta)$ $\theta = 0$ to 4π

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3}{4}\pi$	π	$\frac{5}{4}\pi$	$\frac{3}{2}\pi$	$\frac{7}{4}\pi$	2π
x	0	0.56	0	-1.67	-3.14	-2.78	0	3.89	6.28
y	0	0.56	1.57	1.67	0	-2.78	-4.71	-3.89	0



Test Your Understanding

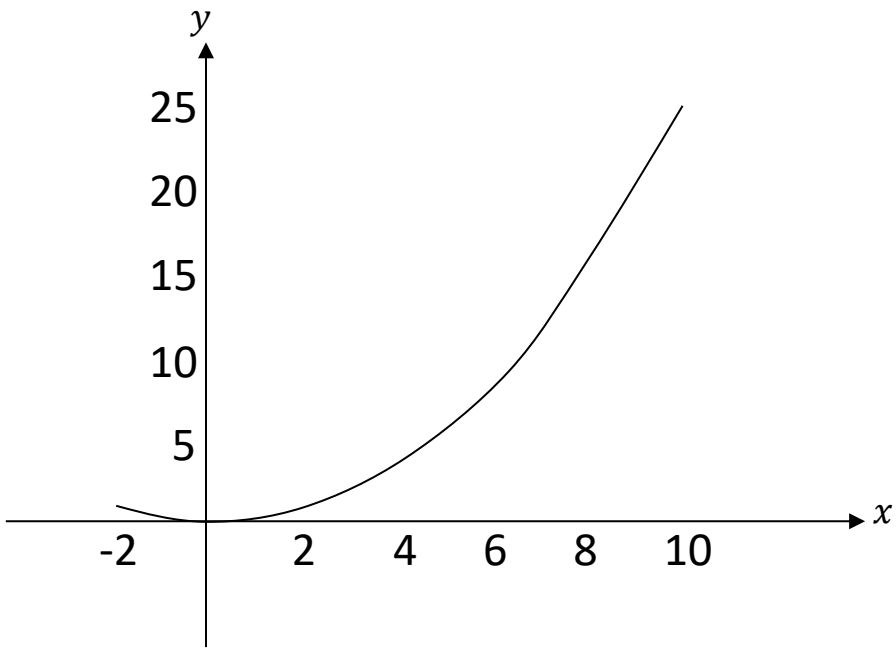
[Textbook] Draw the curve given by the parametric equations $x = 2t$, $y = t^2$, for $-1 \leq t \leq 5$.

?

Test Your Understanding

[Textbook] Draw the curve given by the parametric equations $x = 2t$, $y = t^2$, for $-1 \leq t \leq 5$.

t	-1	0	1	2	3	4	5
x	-2	0	2	4	6	8	10
y	1	0	1	4	9	16	25



Exercise 8.3

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Homework Exercise

1 A curve is given by the parametric equations

$$x = 2t, \quad y = \frac{5}{t}, \quad t \neq 0$$

Copy and complete the table and draw a graph of the curve for $-5 \leqslant t \leqslant 5$.

t	−5	−4	−3	−2	−1	−0.5	0.5	1	2	3	4	5
$x = 2t$	−10	−8				−1						
$y = \frac{5}{t}$	−1	−1.25					10					

2 A curve is given by the parametric equations

$$x = t^2, \quad y = \frac{t^3}{5}$$

Copy and complete the table and draw a graph of the curve for $-4 \leqslant t \leqslant 4$.

t	−4	−3	−2	−1	0	1	2	3	4
$x = t^2$	16								
$y = \frac{t^3}{5}$	−12.8								

Homework Exercise

3 A curve is given by parametric equations

$$x = \tan t + 1, \quad y = \sin t, \quad -\frac{\pi}{4} \leqslant t \leqslant \frac{\pi}{3}$$

Copy and complete the table and draw a graph of the curve for the given domain of t.

t	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	$-\frac{\pi}{12}$	0	$\frac{\pi}{12}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$x = \tan t + 1$	0			1				
$y = \sin t$				0				

4 Sketch the curves given by these parametric equations:

a $x = t - 2, \quad y = t^2 + 1, \quad -4 \leqslant t \leqslant 4$

b $x = 3\sqrt{t}, \quad y = t^3 - 2t, \quad 0 \leqslant t \leqslant 2$

c $x = t^2, \quad y = (2 - t)(t + 3), \quad -5 \leqslant t \leqslant 5$

d $x = 2\sin t - 1, \quad y = 5\cos t + 1, \quad -\frac{\pi}{4} \leqslant t \leqslant \frac{\pi}{4}$

e $x = \sec^2 t - 3, \quad y = 2\sin t + 1, \quad -\frac{\pi}{4} \leqslant t \leqslant \frac{\pi}{2}$

f $x = t - 3\cos t, \quad y = 1 + 2\sin t, \quad 0 \leqslant t \leqslant 2\pi$

Homework Exercise

5 The curve C has parametric equations

$$x = 3 - t, \quad y = t^2 - 2, \quad -2 \leqslant t \leqslant 3$$

a Find a Cartesian equation of C in the form $y = f(x)$. **(4 marks)**

b Sketch the curve C on the appropriate domain. **(3 marks)**

6 The curve C has parametric equations

$$x = 9\cos t - 2, \quad y = 9\sin t + 1, \quad -\frac{\pi}{6} \leqslant t \leqslant \frac{\pi}{2}$$

a Show that the Cartesian equation of C can be written as

$$(x + a)^2 + (y + b)^2 = c$$

where a, b and c are integers to be determined. **(4 marks)**

b Sketch the curve C on the given domain of t. **(3 marks)**

c Find the length of C. **(2 marks)**

Challenge

Sketch the curve given by the parametric equations on the given domain of t:

$$x = \frac{9t}{1 + t^3}, \quad y = \frac{9t^2}{1 + t^3}, \quad t \neq -1$$

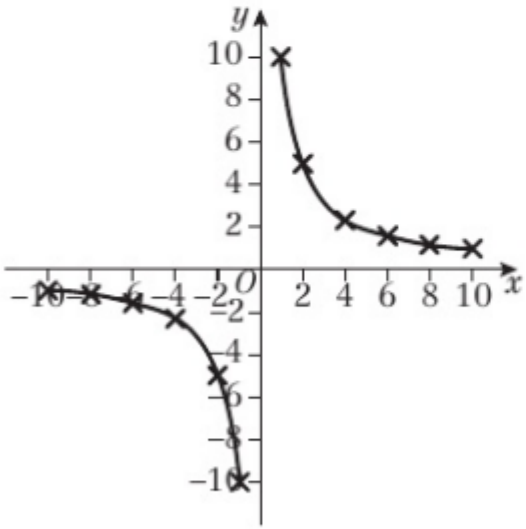
Comment on the behaviour of the curve as t approaches -1 from the positive direction and from the negative direction.

Homework Answers

1

t	−5	−4	−3	−2	−1	−0.5
$x = 2t$	−10	−8	−6	−4	−2	−1
$y = \frac{5}{t}$	−1	−1.25	−1.67	−2.5	−5	−10

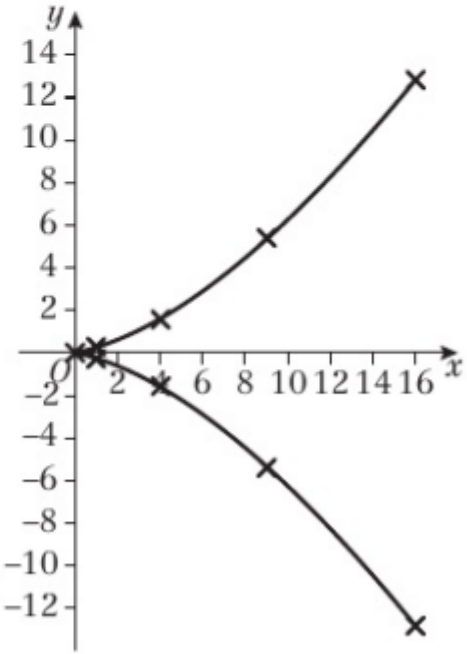
t	0.5	1	2	3	4	5
$x = 2t$	1	2	4	6	8	10
$y = \frac{5}{t}$	10	5	2.5	1.67	1.25	1



2

t	−4	−3	−2	−1	0
$x = t^2$	16	9	4	1	0
$y = \frac{t^3}{5}$	−12.8	−5.4	−1.6	−0.2	0

t	1	2	3	4
$x = t^2$	1	4	9	16
$y = \frac{t^3}{5}$	0.2	1.6	5.4	12.8

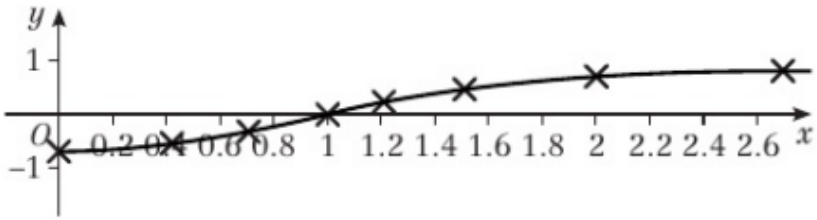


Homework Answers

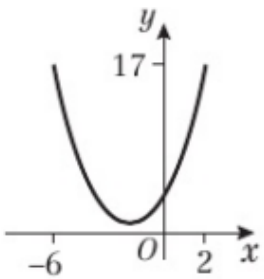
3

t	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	$-\frac{\pi}{12}$	0
$x = \tan t + 1$	0	0.423	0.732	1
$y = \sin t$	−0.707	−0.5	−0.259	0

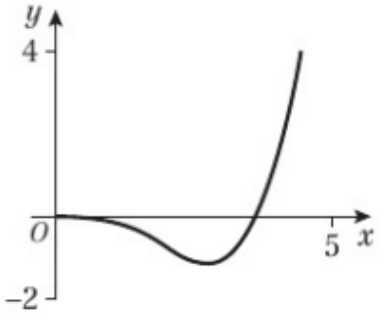
t	$\frac{\pi}{12}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$x = \tan t + 1$	1.268	1.577	2	2.732
$y = \sin t$	0.259	0.5	0.707	0.866



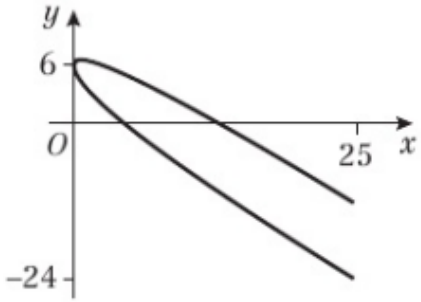
4 **a**



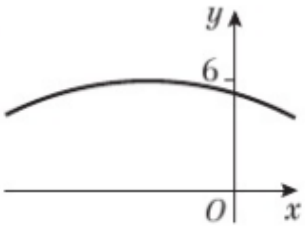
b



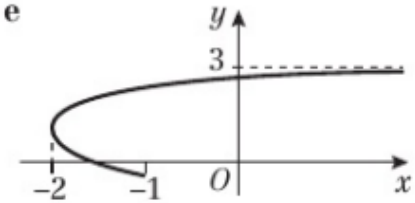
c



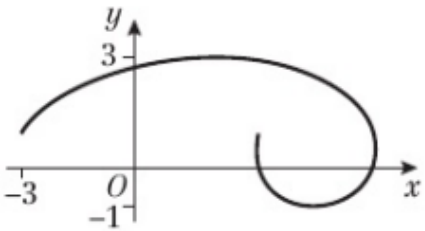
d



e



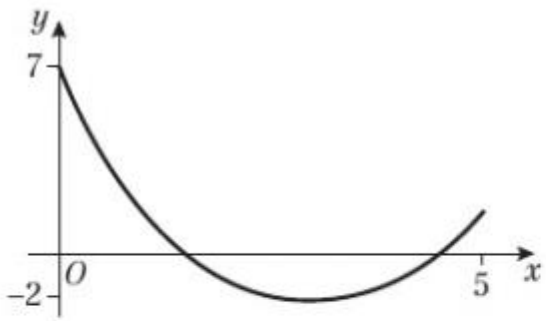
f



Homework Answers

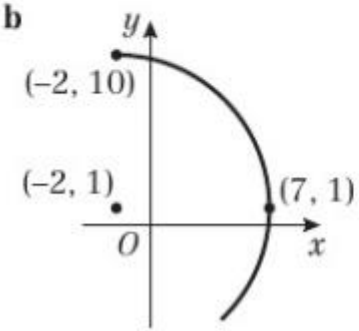
5 **a** $y = (3 - x)^2 - 2$

b



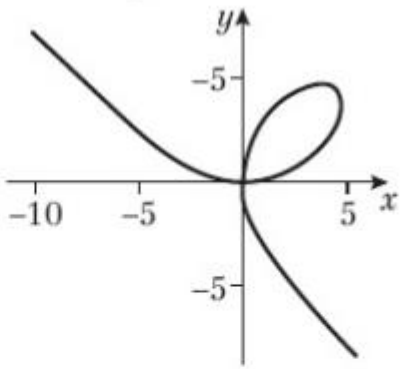
6 **a** $(x + 2)^2 + (y - 1)^2 = 81$

b



c 6π

Challenge



As t approaches −1 from the positive direction, the curve heads off to infinity in the 2nd quadrant.
As t approaches −1 from the negative direction, the curve heads off to infinity in the 4th quadrant.